

VECTOR AUTONOMOUS SYSTEMS

Autonomous VTOL Logistics Infrastructure for Sustainable Urban Delivery

Founder & CEO: Mihail Charilaou

Cyprus | ClimateTech | Applied AI | Autonomous Systems

1. Vision

Vector is building an autonomous aerial logistics network using VTOL cargo drones designed for sustainable middle-mile and last-mile delivery operations.

Our goal is to transform urban logistics in Cyprus by reducing delivery times, traffic dependency, and operational inefficiencies through autonomous drone infrastructure.

2. Problem

Urban delivery systems face growing pressure:

- Traffic congestion increases delivery times
- Courier-based logistics scale poorly
- Operational costs continue to rise
- Tourists and urban residents increasingly expect instant delivery
- Remote and coastal areas remain inefficient to serve

Traditional ground logistics are becoming slower, more expensive, and less scalable.

3. Solution

Vector is developing an autonomous VTOL drone logistics system for:

- Food delivery
- Medicines
- Groceries
- Essential goods
- Time-sensitive urban logistics

By shifting delivery operations from roads to autonomous air infrastructure, Vector enables:

- Faster delivery operations
- Lower long-term operational costs
- Reduced dependence on manual courier labor
- Sustainable urban logistics

4. Technology

Vector combines:

- Autonomous VTOL cargo drones
- AI-assisted routing and fleet management
- Smart station infrastructure
- Automated loading systems
- Real-time logistics optimization

Future development includes exploration of hybrid hydrogen-electric infrastructure for extended-range autonomous logistics and reduced operational emissions.

5. Why Cyprus

Cyprus represents an ideal pilot environment for autonomous logistics:

- Compact urban geography
- High tourism density
- Strong delivery culture
- Short intercity logistics routes
- Favorable climate conditions

- Growing demand for on-demand services

Cyprus can become a strategic launch market for autonomous aerial logistics in the Mediterranean region.

6. Market Opportunity

Cyprus urban delivery market estimate:

€350M–450M annually

Key opportunity areas:

- Food delivery
- Pharmacy logistics
- Convenience commerce
- Tourism-driven instant delivery
- B2B logistics partnerships

Long-term expansion opportunities include:

- Greece
- Israel
- Southern Europe
- Coastal and island cities

7. Business Model

Vector operates a multi-layer revenue model:

- Delivery commissions
- B2B subscriptions
- Logistics partnerships
- Infrastructure leasing
- Fleet operations software
- Branding & advertising

Model:

Infrastructure + Software + Autonomous Operations

8. Competitive Advantage

Vector combines:

- Autonomous aerial logistics
- AI-driven operational optimization
- VTOL flexibility in dense urban environments
- Infrastructure-focused scalability
- Potential hydrogen-electric logistics integration

Unlike traditional delivery services, Vector focuses on building scalable autonomous logistics infrastructure rather than courier-dependent operations.

9. Roadmap

Phase 1 — Pilot & Validation

- MVP and prototype development
- Initial autonomous testing
- Cyprus pilot preparation
- Regulatory pathway research (EASA/SORA)

Goal:

Validate operational feasibility and logistics economics

Phase 2 — City Deployment

- Initial station deployment
- Pilot logistics operations
- Fleet scaling

- AI routing optimization

Goal:

Demonstrate scalable urban operations

Phase 3 — Cyprus Expansion

- Multi-city deployment
- Expanded autonomous operations
- Infrastructure scaling
- Operational profitability targets

Goal:

Establish market leadership in Cyprus

Phase 4 — Regional Expansion & Hydrogen Infrastructure

Target regions:

- Greece
- Israel
- Southern Europe

Expansion strategy:

- Strategic partnerships
- Licensing
- Infrastructure deployment
- Software-driven scaling
- Exploration of hybrid hydrogen-electric logistics infrastructure for extended-range autonomous operations

Long-term vision:

Development of a scalable autonomous aerial logistics network combining AI-driven fleet operations with next-generation sustainable energy infrastructure.

Goal:

Build a regional autonomous logistics platform for coastal and island urban markets.

10. Traction & Development

Current progress:

- VTOL cargo drone architecture in development
- Autonomy system research underway
- Logistics network planning for Cyprus pilot deployment
- Early infrastructure and operational modeling completed
- Market positioning and partnership discussions initiated

11. Team

Mihail Charilaou

Founder & CEO

Background in:

- Robotics
- Software development
- Autonomous systems
- AI-driven infrastructure concepts

Future team expansion:

- Drone engineers
- AI/autonomy specialists
- Operations & compliance
- Mobile and backend systems

12. Investment Opportunity

Vector is currently seeking strategic partners and investors interested in:

- ClimateTech
- Applied AI
- Autonomous systems
- Sustainable logistics infrastructure
- Urban mobility innovation

Target outcome:

Development of scalable autonomous aerial logistics infrastructure for Europe and coastal urban markets.

Projected ROI:

4–10× potential return over 3–5 years under successful regional scaling scenarios.

VECTOR

Building the Future of Autonomous Logistics Infrastructure